

Role of recovery of acetylcholine release in compromised neuromuscular junction function.

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Everyday physical activities, such as walking, are enabled by repeated skeletal muscle contractions and require a well-functioning neuromuscular transmission. In myasthenic disorders, activities of daily living are debilitated by a compromised neuromuscular transmission leading to muscle weakness and fatigability in patients. To enable physical activity, acetylcholine (ACh) is released repeatedly from the motor nerve, however, the role of the nerve terminals' capacity to sustain ACh release to support repetitive contractions under compromised neuromuscular transmission remains unclear. To explore this, we studied synaptic and contractile function during repeated contractions in healthy rat skeletal muscles under conditions of pharmacological induced compromised neuromuscular transmission. Using recordings of endplate potentials, compound muscle action potential (CMAP) and force production in isolated skeletal muscles and living, anesthetized animals, we found that force and CMAP were markedly reduced by even very light activity performed up to 5 s prior to contraction showing that recovery of ACh release was insufficient to maintain synaptic transmission strength. Our results suggest that the timing of depletion and restoration of ACh release may impact clinical signs of weakness and fatigability in patients with impaired neuromuscular transmission and affect the sensitivity of electromyographic recordings in the clinic.

Profiling Novel Quinuclidine-Based Derivatives as Potential Anticholinesterase Drugs: Enzyme Inhibition and Effects on Cell Viability.

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The cholinergic system, relying on the neurotransmitter acetylcholine (ACh), plays a significant role in muscle contraction, cognition, and autonomic nervous system regulation. The enzymes acetylcholinesterase, AChE, and butyrylcholinesterase, BChE, responsible for hydrolyzing ACh, can fine-tune the cholinergic system's activity and are, therefore, excellent pharmacological targets to address a range of medical conditions. We designed, synthesized, and profiled 14 N-alkyl quaternary quinuclidines as inhibitors of human AChE and BChE and analyzed their impact on cell viability to assess their safety in the context of application as potential therapeutics. Our results showed that all of the 14 tested quinuclidines inhibited both AChE and BChE in the micromolar range ($K_i = 0.26 - 156.2 \mu\text{M}$). The highest inhibition potency was observed for two bisquaternary derivatives, 7 (1,1'-(decano)bis(3-hydroxyquinuclidinium bromide)) and 14 (1,1'-(decano)bis(3-hydroxyiminoquinuclidinium bromide)). The cytotoxic effect within 7-200 μM was observed only for monoquaternary quinuclidine derivatives, especially those with the C12-C16 alkyl chain. Further analysis revealed a time-independent mechanism of action, significant LDH release, and a decrease in the cells' mitochondrial membrane potential. Taking all results into consideration, we can confirm that a quinuclidine core presents a good scaffold for cholinesterase binding and that two bisquaternary quinuclidine derivatives could be considered as candidates worth further investigations as drugs acting in the cholinergic system. On the other hand, specific cell-related effects probably triggered by the free long alkyl chain in monoquaternary quinuclidine derivatives should not be neglected in future N-alkyl quaternary quinuclidine derivative structure refinements. Such an effect and their potential to interact with other specific targets, as indicated by a pharmacophore model, open up a new perspective for future investigations of these compounds' scaffold in the treatment of specific conditions and diseases.

other than cholinergic system-linked disorders.

Label-free Imaging of Neurotransmitter Acetylcholine at Neuromuscular Junctions with Stimulated Raman Scattering.

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Acetylcholine is an important neurotransmitter that relays neural excitation from lower motor neurons to muscles. It also plays significant roles in the central nervous system by modulating neurotransmission. However, there is a lack of tools to directly measure the quantity and distribution of acetylcholine at the subcellular level. In this Communication, we demonstrate for the first time that label-free imaging of acetylcholine is achieved with frequency-modulated spectral-focusing stimulated Raman scattering (FMSF-SRS) microscopy: a technical improvement over traditional SRS microscopy that effectively removes imaging backgrounds. Moreover, we directly quantified the local concentration of acetylcholine at the neuromuscular junction of frog cutaneous pectoris muscle.

Measuring *Caenorhabditis elegans* Sensitivity to the Acetylcholine Receptor Agonist Levamisole.

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At the neuromuscular junction (NMJ), the binding of the excitatory neurotransmitter acetylcholine (ACh) to postsynaptic receptors leads to muscle contraction. As in vertebrate skeletal muscle, cholinergic signaling in the body wall muscles of the model organism *Caenorhabditis elegans* is required for locomotion. Exposure to levamisole, a pharmacological agonist of one class of ACh receptors on the body wall muscles, causes time-dependent paralysis of wild-type animals. Altered sensitivity to levamisole suggests defects in signaling at the NMJ or muscle function. Here, a protocol for a liquid levamisole assay performed on *C. elegans* grown in 24-well plates is presented. Vigorous swimming of the animals in liquid allows for the assessment and quantitation of levamisole-induced paralysis in hundreds of worms over a one-hour time period without requiring physical manipulation. This procedure can be used with both wild-type and mutants that have altered sensitivity to levamisole to demonstrate the functional consequences of altered signaling at the NMJ.

Structural mechanism of muscle nicotinic receptor desensitization and block by curare.

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Binding of the neurotransmitter acetylcholine to its receptors on muscle fibers depolarizes the membrane and thereby triggers muscle contraction. We sought to understand at the level of three-dimensional structure how agonists and antagonists alter nicotinic acetylcholine receptor conformation. We used the muscle-type receptor from the Torpedo ray to first define the structure of the receptor in a resting, activatable state. We then determined the receptor structure bound to the agonist carbachol, which stabilizes an asymmetric, closed channel desensitized state. We find conformational changes in a peripheral membrane helix are tied to recovery from desensitization. To probe mechanisms of

antagonism, we obtained receptor structures with the active component of curare, a poison arrow toxin and precursor to modern muscle relaxants. d-Tubocurarine stabilizes the receptor in a desensitized-like state in the presence and absence of agonist. These findings define the transitions between resting and desensitized states and reveal divergent means by which antagonists block channel activity of the muscle-type nicotinic receptor.

Nicotinic acetylcholine receptor redux: Discovery of accessories opens therapeutic vistas.

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The neurotransmitter acetylcholine (ACh) acts in part through a family of nicotinic ACh receptors (nAChRs), which mediate diverse physiological processes including muscle contraction, neurotransmission, and sensory transduction. Pharmacologically, nAChRs are responsible for tobacco addiction and are targeted by medicines for hypertension and dementia. Nicotinic AChRs were the first ion channels to be isolated. Recent studies have identified molecules that control nAChR biogenesis, trafficking, and function. These nAChR accessories include protein and chemical chaperones as well as auxiliary subunits. Whereas some factors act on many nAChRs, others are receptor specific. Discovery of these regulatory mechanisms is transforming nAChR research in cells and tissues ranging from central neurons to spinal ganglia to cochlear hair cells. Nicotinic AChR-specific accessories also enable drug discovery on high-confidence targets for psychiatric, neurological, and auditory disorders.

Cholinesterases in Tripartite Neuromuscular Synapse.

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The neuromuscular junction (NMJ) is a tripartite synapse in which not only presynaptic and post-synaptic cells participate in synaptic transmission, but also terminal Schwann cells (TSC). Acetylcholine (ACh) is the neurotransmitter that mediates the signal between the motor neuron and the muscle but also between the motor neuron and TSC. ACh action is terminated by acetylcholinesterase (AChE), anchored by collagen Q (ColQ) in the basal lamina of NMJs. AChE is also anchored by a proline-rich membrane anchor (PRiMA) to the surface of the nerve terminal. Butyrylcholinesterase (BChE), a second cholinesterase, is abundant on TSC and anchored by PRiMA to its plasma membrane. Genetic studies in mice have revealed different regulations of synaptic transmission that depend on ACh spillover. One of the strongest is a depression of ACh release that depends on the activation of $\alpha 7$ nicotinic acetylcholine receptors (nAChR). Partial AChE deficiency has been described in many pathologies or during treatment with cholinesterase inhibitors. In addition to changing the activation of muscle nAChR, AChE deficiency results in an ACh spillover that changes TSC signaling. In this mini-review, we will first briefly outline the organization of the NMJ. This will be followed by a look at the role of TSC in synaptic transmission. Finally, we will review the pathological conditions where there is evidence of decreased AChE activity.

Effects of cholinergic drugs on muscle contraction in *Moniliformis moniliformis* (Acanthocephala).

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In whole *Moniliformis moniliformis* spontaneous muscle contractions were rhythmic; longitudinal contractions were measured with a force transducer. The cholinergic agonists levamisole and nicotine significantly increased muscle tension in whole worms; these contractions were tonic and were antagonised by the ganglionic blocker pentolinium and by piperazine. In addition, levamisole-induced contractions were inhibited by gallamine, hexamethonium, and norepinephrine. In worm segments, where drugs in solution were injected through the worms, acetylcholine (ACh) and nicotinic agonists were effective in causing contractions, whereas muscarinic agonists in concentrations up to 1 mM had no effect. Although muscle contraction in *M. moniliformis* was induced by nicotinic agonists, these contractions were effectively antagonised by a range of chemicals that block ganglionic, skeletal, and muscarinic sites in vertebrates. The presence of ACh in *M. moniliformis* and the effects of nicotinic agonists on muscle contraction suggest that ACh is a putative excitatory neurotransmitter.

Neural control of gastro-intestinal motility; events following receptor activation.

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The motility of the mammalian gastro-intestinal tract is regulated by a number of neuronal systems. The postganglionic, intramural parts of these systems are characterized by their neurotransmitters as cholinergic, adrenergic and 'purinergic' nerves, respectively. The transmitter acetylcholine evokes contractions of the smooth muscle cell, whereas both noradrenalin and the transmitter from the 'purinergic' nerves, presumably ATP, induce relaxations of smooth muscles. Agents and actions which influence the neurohumoral transmission and the mechanisms underlying the effects induced by the transmitters in smooth muscle cells are reviewed.

Gastrointestinal neurotransmitters.

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The enteric nervous system contains neurones that are intrinsic to the gastrointestinal tract and the axons of extrinsic neurones. More than 30 functional types of neurone are present and about 25 different possible neurotransmitters have been identified in enteric neurones. Most neurones utilize several transmitters; amongst the transmitters of an individual neurone, one is usually a primary transmitter and other substances are subsidiary transmitters or neuromodulators. The primary transmitter is the substance that has the major role in acutely changing the excitability of the innervated cell. Current evidence indicates that primary transmitters are strongly conserved; that is, the same substance will be the neurotransmitter in functionally equivalent neurones in different regions of the gastrointestinal tract and in different species. In contrast, subsidiary transmitters and neuromodulators of equivalent neurones in different regions are not necessarily the same. Only about seven of the approximately 25 enteric neurotransmitters are known to be primary transmitters. Acetylcholine is the primary transmitter of vagal and pelvic preganglionic neurones, of enteric interneurones, of one class of secretomotor neurone in the intestine and of motor neurones controlling gastric acid secretion. Acetylcholine and tachykinins are co-primary transmitters of muscle motor neurones, with acetylcholine appearing to have the greater role. Tachykinins are probably primary transmitters of enteric sensory neurones at neuroneuronal synapses. Serotonin may also be a transmitter to neurones in the enteric ganglia. Nitric oxide appears to be the

usual primary transmitter of enteric inhibitory motor neurones to the muscle. ATP and vasoactive intestinal peptide are subsidiary transmitters of these neurones, although in some regions they may have a primary transmitter role. Vasoactive intestinal peptide is the primary transmitter of non-cholinergic secretomotor neurones. Gastrin releasing peptide is the primary transmitter of motor neurones to gastrin cells. Noradrenaline is the primary transmitter of sympathetic neurones that supply the intestine.
